Experiment-7

Convolution with Continuous and Discrete Signals.

This experiment is intended to make the student to use MATLAB for experiments, convolution operations on continuous and discrete time signals. It is expected that the student will write a "readable" code in a file and execute. Use only Editor Window to write the Matlab Code.

The student is expected to be conversant with Matlab commands: axis, text, grid, hold on, hold off, clc, clear all, close all, zeros, ones, etc.

In all these experiments, the time scale (x-axis) should have a resolution of 1 msec. Use Example 1 for creating such time axis.

%----------Example 1---------- %

Time axis for signals dt = 0.001;

% Time axis resolution 1 msec Stime = -5 ;

% Start Time of the time axis. In this example it is -5 seconds Etime = 5;

% End time of Time axis.

In this example it is 5 seconds Mytime = [Stime:dt:Etime]; % Total time axis % You may change the Stime and Etime as per the need.

Further, usually we may intend to define or display a signal over certain time duration, say between t1 and t2 seconds even though the signal may exists beyond these limits.

Convolution involves the following operations.

1. Folding
2. Multiplication
3. Addition
4. Shifting

These operations can be represented by a Mathematical Expression as follows:

x[n]= Input signal Samples

h[n-k]= Impulse response co-efficient.

y[n]= Convolution output.

n = No. of Input samples

h = No. of Impulse response co-efficient.

Example : X(n)={1 2 -1 0 1}, h(n)={ 1,2,3,-1}.

PROBLEM STATEMENTS:-

% _____%

A. Convolution of two different continuous signals using built- in function:

1. Write a matlab program to perform convolution between sinusoidal and square function and see how matlab reacts to it.
2. Write a matlab program to perform convolution between sinusoidal and square function without using Built-In function and see how matlab reacts to it.

% _____%

B. Convolution of two different sequences using built- in function:

1. Write a matlab program to perform the convolution between two different user define sequences.

% _____%

C. Convolution of two different sequences without using built- in function:

1. Write a matlab program to perform the convolution between two different user define sequences without using Built-In function.

% _____%